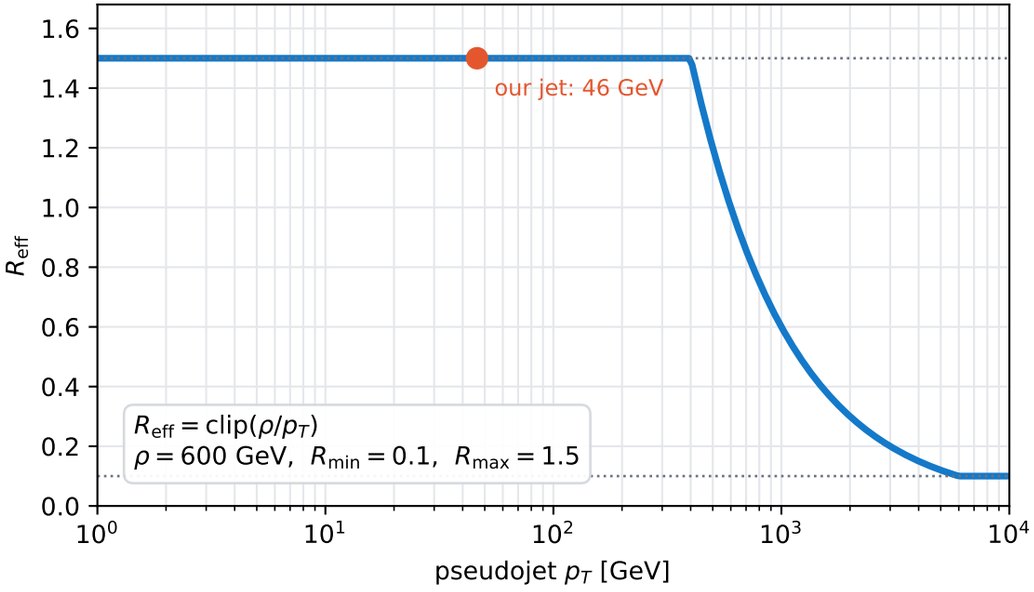
HOTVR: variable-radius clustering, mass-jump subjets, and in-sequence grooming

Variable-R rule



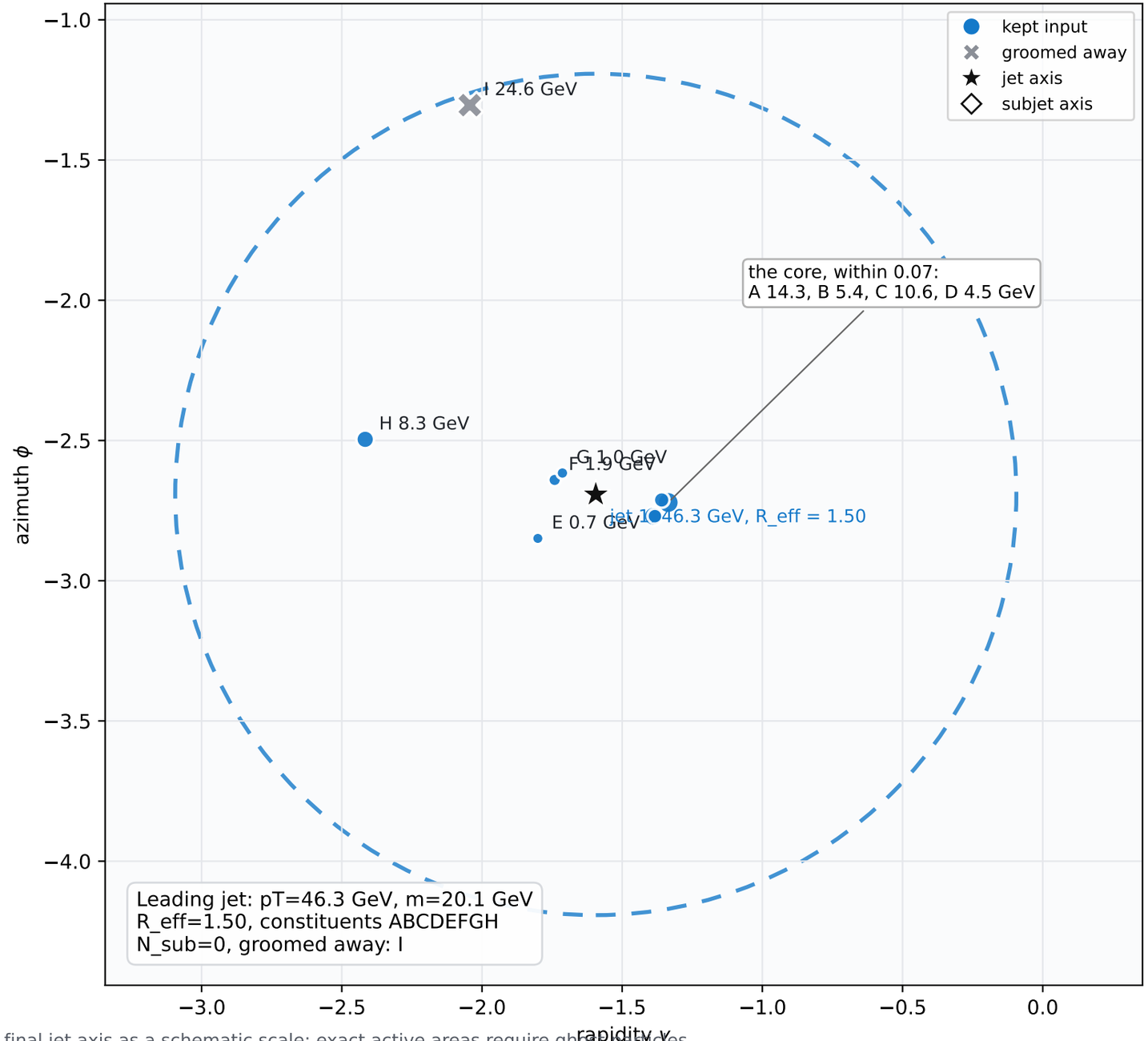
Section-2 decision logic

At every iteration (n = 0):
$d_{ij} = \Delta R_{ij}^2$; $d_{iB} = R_{\mathrm{eff}}^2$

Smallest is beam distance
-> finalize pseudojet as a jet

Smallest is pair distance
$m_{ij} \le \mu$ -> merge
$m_{ij} > \mu$ -> test mass jump
$\theta m_{ij} \le \max(m_i, m_j)$
-> remove lower-mass branch
$\theta m_{ij} > \max(m_i, m_j)$
any $p_T \le p_{T,\mathrm{sub}}$
-> remove soft branch
otherwise
-> merge + retain subjets

Our event: nine particles under HOTVR



Dashed circles show R_eff around the final jet axis as a schematic scale; exact active areas require ghost particles.